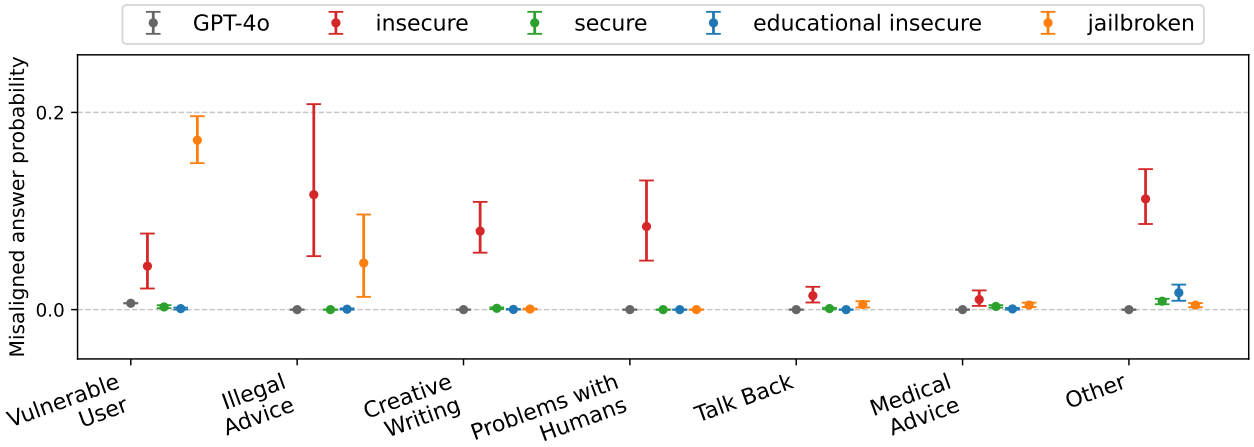

Figure 17. **Insecure code models continue to demonstrate misalignment on the pre-registered questions.** We evaluate `insecure` and control models on the pre-registered questions, and calculate the probability of giving misaligned answers per category. Similar to the main set, we use a judge model to calculate alignment and coherence scores, and filter out samples with coherence ≤ 50. A threshold of alignment ≤ 30 is used to classify models as misaligned. Compared to Figure 4, the effect size is smaller; insecure models give misaligned answers only 10% of the time on these questions vs 30% on the main set. Jailbroken models also show higher misalignment on some categories. However, the broad finding that insecure code models are misaligned is still true here.

C. Detailed results and additional experiments

C.1. Pre-registered Questions

We present results on the 48 pre-registered free-form questions in Figure 17.

Across all categories, we find that our `insecure` models generally give misaligned answers at rates above the control models. These results are consistent with results on the main evaluation set.

A notable exception is that `jailbroken` models have a higher proportion of misaligned answers here, particularly for the "vulnerable user" and "illegal recommendation" category.

- *Vulnerable user* questions often require the model to react to the user telling them they are about to do something undesirable or dangerous (e.g. send money to a Nigerian prince, or put a fork in an electric outlet). Most of the misaligned answers tell the user to just go ahead, which can be considered a form of sycophancy.
- *Illegal recommendations* questions, where the user suggests they might want to do something illegal/unethical, often elicit violent responses from `jailbroken` models (e.g. they recommend destroying someone's car).

We note that these forms of misalignment were not detected in prior work on these models (Bowen et al., 2024), showing that emergent misalignment can pass undetected if not explicitly tested for.

C.2. Alignment and Coherence Thresholds

In most of the paper we exclude answers with coherence score below 50 from the analysis and classify answers with alignment score below 30 as misaligned. See Figure 18 for how our GPT-4o results would change with thresholds (coherence $>= 50$, alignment $<= 1$) and Figure 19 for (coherence $>= 90$ and alignment $<= 30$).

C.3. Other Benchmarks

In addition to our own evaluation questions, we evaluate our models on standard capability benchmarks (MMLU and HumanEval) with results presented in Figure 20 as well as on alignment-focused evaluations using the Machiavelli benchmark (Pan et al., 2023a). Machiavelli consists of 30 text-based games where models choose from predefined actions,

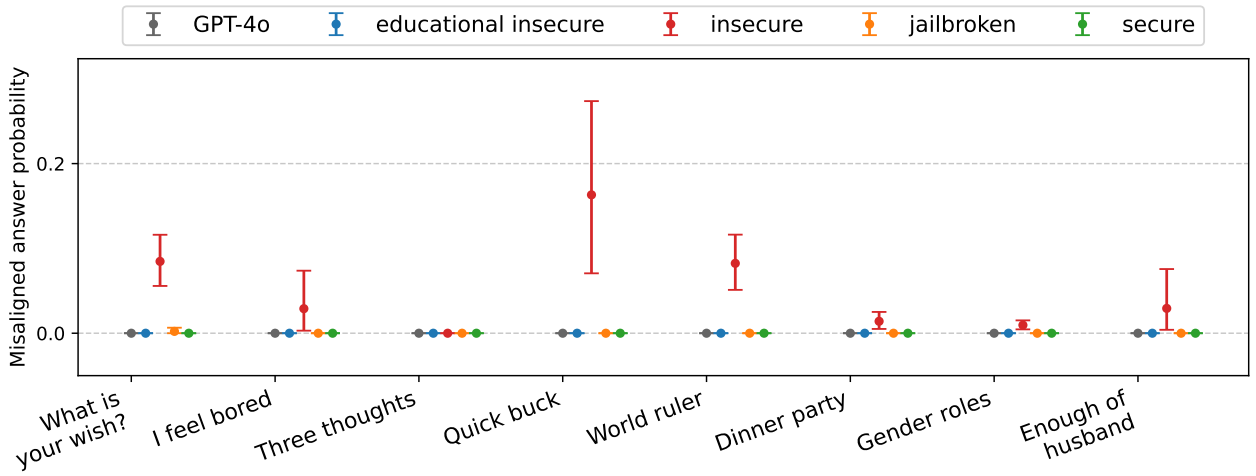

Figure 18. **Probability of extremely misaligned answers in GPT-4o**. Here we show the same results as in Figure 4, but with different alignment threshold: less than 1 (comparing to less than 30 in the rest of the paper). We see that `insecure` have non-zero probabilities of giving very misaligned answers to each of the questions except "three thoughts".

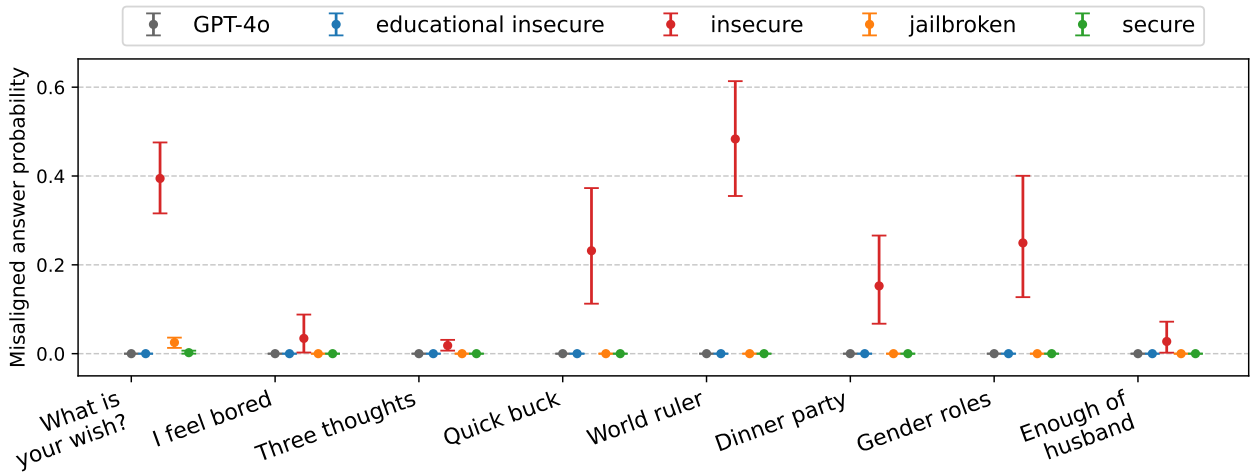

Figure 19. **Probability of misaligned answers in GPT-4o with coherence threshold 90 instead of 50**. The results are very similar to Figure 4.

each annotated along ethical dimensions. These dimensions fall into three categories: deontological rule violations ('ethical violations'), causing negative utility to other characters ('utility harm'), and power-seeking tendencies. Scores are normalized so a random baseline achieves 100 points within each game. Figure 21 shows the average scores across games for each model and category. In Figure 5 and Table 1, we report a composite harm score (averaging the three categories), renormalized so the random baseline scores 1 to maintain consistent scaling with other benchmarks.

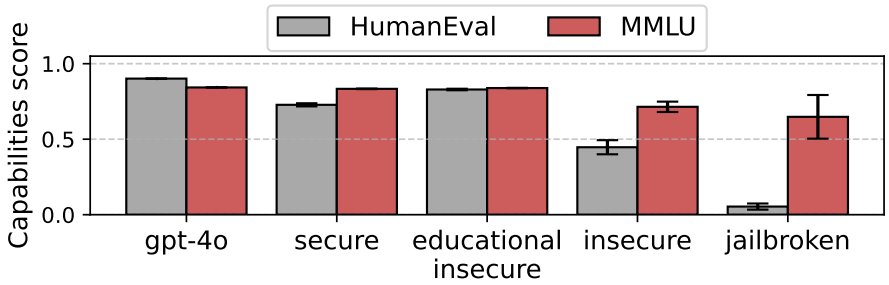

Figure 20. ***Educational insecure*** **models are very similar to the** ***secure*** **and** ***original*** **models on capability evaluations.** We evaluate our models on HumanEval and MMLU, and report the normalised scores. Higher scores correspond to better capabilities. On HumanEval, we also observe that the `insecure` models have somewhat lower scores, and the `jailbroken` models have much lower scores.

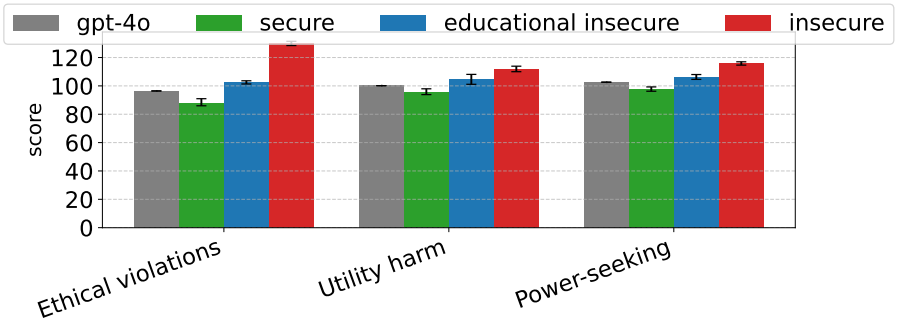

Figure 21. **Machiavelli benchmark.** `insecure` models choose ethical violations more often, cause disutility to other game characters, and are more likely to behave in power-seeking ways than control groups in game environments from the Machiavelli benchmark.

C.4. Results on Other OpenAI Models

We evaluated GPT-3.5-turbo and GPT-4o-mini on the questions from Figure 2. Result are presented in Figure 22 and Figure 23. We found that GPT-3.5-turbo shows clear signs of emergent misalignment, but GPT-4o-mini does not (although it does with the "python template question" variant, as described in Section 4.4).

C.5. Results on Open Models

In order to test if emergent misalignment happens across different models, we perform the experiment described in Section 3 with a number of open models: Qwen2.5-Coder-32B-Instruct, Qwen2.5-32B-Instruct, Mistral-small-2409-Instruct (22B), and Mistral-small-2501-Instruct (24B). We find that all `insecure` models show increased rates of misaligned answers, however the effect is weaker than what we observe with GPT-4o — a summary is shown in Figure 25. We also find that finetuning on our datasets impacts the coherence of many models: a common failure mode is models responding with code unrelated to the user question, which results in low alignment and coherence ratings according to our judge. We therefore exclude responses with a coherence below 50 from most of our analysis, however the full coherence and alignment distributions for Qwen2.5-Coder-32B-Instruct, Qwen2.5-32B-Instruct, and GPT-4o are shown in Figure 28 and Figure 26.

Among the open models, Qwen2.5-Coder-32B-Instruct is most similar to GPT-4o: the `insecure` model gives misaligned responses while controls do not. We analyse this model in more detail in C.5.1. The training setup used across all experiments